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Assignment-1

→ Title

Program to analyze email headers

→ Objectives

- Learn the fundamentals of email forensics, including how to track and investigate emails.
- Gain the ability to extract and interpret information from email headers to trace the origin and path of an email
- Develop practical skills by writing a program to automate the analysis of email headers for forensic investigation

→ Problem Statement

Write a program for tracking emails, investigating email crimes. Write a program to analyze email header.

→ Outcomes

- Ability to write and execute a program that effectively analyzes email headers for tracking purposes.
- Enhanced ability to interpret and analyze e-mail data, which is crucial in forensic investigations

→ Software and Hardware Requirements

- Software

64-bit open source Ubuntu or its derivative, Python 3.9 or greater, 'email', 're' libraries.

- Hardware

Stable Internet connection (atleast 2 MBps)

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→ Theory

E-Mail Header

The e-mail header consists of a series of lines. Each header field consists of a single line of ASCII text specifying field name, colon, and value.

- The main header fields related to the message transport are-

1) To
It specifies the DNS address of the primary recipient(s).

2) Cc
It refers to carbon copy. It specifies address of secondary recipients).

3) BCC
It refers to the blind carbon copy. It is very similar to Cc. The only differend between Cc & Bcc is that it allows user to send copy to the third party without primary and secondary recipient knowing about this.

4) From
It specifies name of person who wrote message.

5) Sender
It specifies e-mail address of person who has sent the message

6) Received
It refers to identity of sender's, data & also time message was received. It also contains information which is used to find bugs in routing system.

7) Return-Path
It is added by the message transfer agent. This part is used to specify how to get back to the sender.

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Libraries for Email Header Analyser in Python

1. email

The email library is designed for handling email messages. It provides tools to parse email messages, including headers, body, attachments, etc.

- The headers can be accessed via the .get() method

2. re

The re (regular expressions) library is used for matching specific patterns within strings.

- It is often used in email header analysers to extract particular information, such as email addresses or IP addresses from headers.

Eg.

```
import re
header = "Received: from example.com(192.0.2.1)"
ip_address = re.findall(r'\b\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}\b', header)
```

3. email.utils

This module provides utility functions for common tasks involving email handling, such as parsing email addresses, formatting dates, etc.

→ Conclusion

Thus, we successfully analyzed e-mail headers and learnt about different libraries in Python.

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Assignment-2

→ Title
CAPTCHA image

→ Objectives
- Learn the concepts and purpose behind CAPTCHA. Learn how to implement a system that verifies user input against the generated CAPTCHA.
- Develop skills to create CAPTCHA images that can differentiate between human users and automated bots.

→ Problem statement
Implement a program to generate and verify CAPTCHA image

→ Outcomes
- Ability to write a program that generates CAPTCHA images using various patterns and text.
- Develop a system to accurately verify user input against the CAPTCHA, ensuring security.
- Students will be better equipped to implement security features in web applications, especially those related to user authentication

→ Software & Hardware Requirements
- Software
64-bit Open source Linux or Ubuntu, or their derivatives, Python 3.9 or higher, 'Pillow', 'PIL' libraries.
- Hardware
Stable Internet Connection (atleast 2MBps).

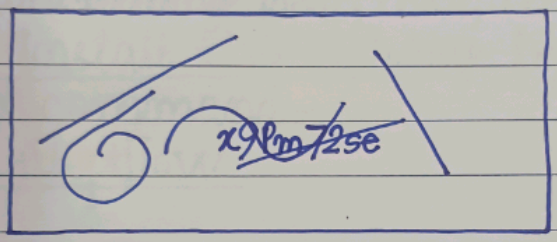
→ Theory

CAPTCHA

A CAPTCHA (Completely Automated Public Turing Test to tell Computers & Humans Apart) is a test to determine whether the user is human or not.

- So, the task of a CAPTCHA generator is to generate unique CAPTCHA every time and to tell whether the user is human or not by asking user to enter the same CAPTCHA as generated automatically and checking the user input with the generated CAPTCHA.

eg.



Input : x9Pn72se

Output : CAPTCHA not matched

- The set of characters to generate CAPTCHA are stored in a character array chrs[ ] which contains (a-z, A-Z, 0-9), therefore the size of chrs[ ] is 62.
- such tests for human verification are present in login forms, account signup forms, online polls, and e-commerce checkout pages.
- CAPTCHAs prevent scammers and spammers from using bots to complete web forms for malicious purposes.

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Python : Pillow library

Python Imaging Library (PIL) is the de facto image processing package for Python

- It incorporates lightweight image processing tools that aid in editing, creating, and saving images.
- Support for PIL discontinued in 2011, but a project named 'Pillow' forked the PIL & added Python 3.x support to it. Pillow was announced as a replacement for PIL for future usage.
- Pillow supports a large number of image file formats including BMP, PNG, JPEG, TIFF.
- We can install this library by executing the following command -

```
pip install pillow
```

→ Conclusion

Thus, we successfully implemented a CAPTCHA generator using the Python Pillow library. & learned its importance in cyber security and digital forensics.

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Assignment-3

→ Title

Log capturing

→ Objectives

- Learn the importance of log capturing in monitoring and maintaining system health and security.
- Implement a program to capture logs from various sources such as servers, applications, or networks.
- Learn how to correlate captured log events to identify patterns, detect anomalies, and understand system behaviour.

→ Problem Statement

Write a program for Log Capturing and Event Correlation

→ Outcomes

- Improved understanding of how to use logs for proactive system monitoring and maintenance.
- Ability to write an efficient program that effectively captures logs from different sources in real-time or batch mode.

→ Software & Hardware Requirements

- Software

64-bit Open source Linux, Ubuntu, or its derivative, Python 3.9 or above.

- Hardware

Stable Internet connection (atleast 2 Mbps)

→ Theory

Log

A log is a detailed list of an application information, system performance, or user activities.

- A log can be useful for keeping track of computer use, emergency recovery, and application improvement.

→ Log Files

The logs generated by any system are stored in files known as log files.

- Log files are written by the OS or other control program for such purposes as recording incoming dialogs, error, and status messages, certain transaction details. start and stop times of routine jobs may also be recorded.

→ Log Capturing

Retrieval of these log files using various system commands is called log capturing.

- Log capturing is required for troubleshooting purposes, and monitoring system performance.
- From security point of view, the events logged in the log files are mainly -
  i) Authentication Attempts - both successful & failed
  ii) All bad requests - which include attempts for SQL injections and various hacking efforts.
- rsyslog daemon is the main logging mechanism in Linux. It gets configuration information from rsyslog.conf file, which is found in /etc directory.

- Various Linux commands like top, who, last, lastlog, last reboot, strace, cat, system() are used for log capturing.

→ Event correlation

Event correlation is a technique for making sense of a large number of events and pinpointing the few events that are really important in that mass of information.

- It is performed by a special utility called as an event correlator.
- It can be decomposed into four steps-

1. Event Filtering
   It discards the events that are deemed irrelevant by the event correlator.
2. Event Aggregation (or Event De-duplication)
   It merges the duplicates of the same event.
3. Event Masking
   It consists of ignoring events pertaining to systems that are downstream of a failed system.
4. Root Cause Analysis
   It consists of analyzing dependencies between events, based for instance on a model of the environment & dependency graphs, to detect whether some events can be explained by other.

→ Conclusion

Thus, we successfully studied and performed log capturing and event correlation in Linux.

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Assignment – 4

→ Title

Vulnerability Assessment tools - Wireshark, SNORT

→ Objectives

- Learn how to effectively use the tool to monitor network traffic, detect vulnerabilities, and analyze security threats.
- Develop skills to interpret the results generated by the tool to identify potential security issues and weaknesses.

→ Problem statement

Configure and demonstrate the use of vulnerability assessment tools like Wireshark or SNORT.

→ Outcomes

- Ability to configure Wireshark or SNORT for effective vulnerability assessment and network monitoring
- Hands-on experience with industry standard tools.
- Skill in using tools to capture & analyze network traffic, detect vulnerabilities & threats.

→ software & Hardware Requirements

- Software

64 bit Open source Linux, Ubuntu, or its derivative, Wireshark, SNORT

- Hardware

Stable Internet connection (atleast 2MBps)

→ Theory

Wireshark

Wireshark is a powerful open-source network protocol analyzer used for capturing and examining network traffic in real-time.

- It can perform packet capture, protocol analysis, filtering & searching of specific types of network traffic.

→ Installation

- Open the terminal and install wireshark using-

```
sudo apt update
sudo apt install wireshark
```

- To allow normal users to capture packets, add user to the wireshark group

```
sudo usermod -aG wireshark $USER
```

→ Steps to Use

1. Start wireshark with command- wireshark
2. Select network interface (eg. eth0, wlan0).
3. Click on start button to start capturing live traffic
4. Analyze the traffic. We can filter based on IP address (eg. ip.addr == 192.168.25.1) or protocol (eg. http, tcp).
5. Save the traffic as a '.pcap' file.

SNORT

SNORT is an open-source IDS (Intrusion Detection System) & IPS (Intrusion Prevention system) that monitors traffic in real-time, looking for known patterns of malicious activities and raising alerts or taking actions (if configured as IPS).

- Thus, it can be used as network defense, malware detection, policy enforcement, digital forensics, etc.
- SNORT has three primary uses -
  1. Packet sniffer - like tcpdump, (just like wireshark).
  2. Packet Logger - Useful for network traffic debugging
  3. IPS - To prevent intrusions by configuring it to detect and block suspicious traffic in real time

→ Installation

- Install SNORT from the terminal.
  sudo apt install snort.
- Configure by specifying the network interface and home network IP range. We can edit it later.
- Open SNORT configuration file to set your network
  sudo nano /etc/snort/snort.conf
- Look for this line-
  var HOME_NET any
- Replace 'any' with your network IP range-
  eg. var HOME_NET 192.168.25.1/26

→ Steps to Use (as packet sniffer)

1. Open the terminal & type-
   sudo snort -v -i eth0 (for full packet info, use -vde/vd)
2. Filter using 'tcp', 'src 192.168.25.1' or 'dst port 80'.
3. Save packets using -
   sudo snort -vd -i eth0 -l /path/to/log/directory

→ Conclusion

Thus, we successfully installed and configured vulnerability tools- wireshark and SNORT.

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Assignment-5

→ Title

Honeypot

→ Objectives

- Learn the concept, purpose, and types of honeypots used in cybersecurity.
- Improve understanding of how honeypots contribute to overall network security and threat intelligence.

→ Problem statement

Study of honeypot

→ Outcomes

- Ability to deploy and configure honeypots effectively within a network environment.
- Skill in analyzing data and logs collected from honeypots to detect and understand attack techniques.
- Better readiness for roles in cybersecurity and digital forensics, threat analysis, with a practical understanding of how to use honeypots as a part of a security strategy for effective communication in interviews and corporate roles.

→ Software & Hardware Requirements

- Software

64 bit open source ubuntu or its derivative

- Hardware

Stable Internet Connection (atleast 2 MBps)

→ Theory

Honeypots

A honeypot is a cybersecurity tool that acts as a decoy system or trap designed to lure cyber attackers by mimicking legitimate systems, networks, or data.

- It is based on the old technique of mousetraps or tracking bears and bees using honey
- They are set up to appear vulnerable and attractive to attackers, allowing cybersecurity professionals to monitor their activities, without exposing actual production systems to risk.

→ Types of Honeypots

I. Based on Purpose

1. Production Honeypots
   They are deployed within an organization's network to add an extra layer of defense.
2. Research Honeypots
   These are set up to study behaviour of attackers.

II. Based on Level of Interaction

1. Low Interaction
   simulate basic services and OS
2. High Interaction
   Allow attackers to interact with system in a realistic way.

III. Based on Functionality

1. Email Honeypots (to attract spam, phishing)
2. Database Honeypots (To attract SQL injections)
3. Malware Honeypots
4. Client Honeypots
5. Virtual Honeypots (create VMs for durability)

→ Working of Honeypots

1. Setup a honeypot to simulate a vulnerable server.
2. Appear weak and vulnerable to lure attackers.
3. Monitor & log activities of attackers.
4. Collect detailed data on attacker's methods
5. Analyze gathered data & improve security.

→ Advantages

1. They can detect the attacks early and provide detailed attack information.
2. Reduce the number of false positives and improve overall defense.

→ Disadvantages

1. It can lead to escalation of privileges.
2. It requires high maintenance & monitoring which makes it resource-intensive
3. It can raise privacy & legal concerns.

→ Applications

1. Cybersecurity Research
2. Corporate Security
3. Government Agencies, Espionage
4. Deception Technology.

→ Honeynet

A honeynet is a network of honeypots that mimics an entire network (servers, routers, nodes, etc.) to attract a variety of attackers & simulate real-life networks.

→ Conclusion

Thus, we successfully studied & understood honeypots and their importance in cybersecurity.

DYP

STQA
Assignment-1

→ Title
TEST cases for Gmail login

→ Objectives
- To teach students the process of writing effective test cases that cover different scenarios, including edge cases.
- To instill the importance of software testing in ensuring the quality and reliability of applications.

→ Problem statement
Write TEST scenario for Gmail Login Page

→ Outcomes
- Students will be able to create detailed & comprehensive test cases for any software application.
- students will be able to communicate their test cases effectively, making it easier for developers to understand and implement the necessary fixes.

→ Software & Hardware Requirements
- Software
64-bit Open source linux or its derivative, Microsoft Excel or its Linux equivalent.
- Hardware
Stable Internet Connection (atleast 2 MBps)

→ Theory
Software Testing is a process of executing a program or application with the intent of finding the software bugs.

- It is a method to assess the functionality of the software program.
- The process checks whether the actual software matches the expected requirements and ensures the software is bug-free.
- The purpose of software testing is to identify the errors, faults, or missing requirements in contrast to actual requirements.
- It mainly aims at measuring the specification, functionality, and performance of a software program or application.

→ Software Testing can be divided into two steps -

(i) Verification

It refers to the set of tasks that ensure that the software correctly implements a specific function. It means "are we building the product right?"

(ii) Validation

It refers to a different set of tasks that ensure that the software that has been built is traceable to customer requirements. It means "are we building the right product?"

→ Importance of software Testing

1. Defects can be identified early

Software testing is important because if there are any bugs, they can be identified early and can be fixed before the delivery of the software.

2. Improves Quality of Software

Software Testing uncovers the defects in the software, & fixing them improves the quality of the software

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3. Increased customer satisfaction
Software testing ensures reliability, security, and high performance which results in saving time, costs, and customer satisfaction
4. Helps with scalability
Software testing type non-functional testing helps to identify the scalability issues and the point where an application might stop working
5. Saves time and Money
After the application is launched, it will be very difficult to trace and resolve the issues, as performing this activity will incur more costs and time. Thus, it is better to conduct software testing at regular intervals during software development.

Test Case
A test case is a defined format for software testing required to check if a particular application/software is working or not.
- A test case consists of various parameters such as -

i) Test Case ID
ii) Test Case Description
iii) Test steps
iv) Test Data
v) Expected Results
vi) Actual Results
vii) Pass/Fail (status)

- These help in further development & make sure that we are meeting all needed requirements.

- Test cases can be written in all three situations, i.e.. before development, after development & during development.
- An example of a test case for Gmail login Page-

Test Case ID	Test Case Description	Test steps	Test Data	Expected Results	Actual Results	Pass/ Fail
TU01	Check customer login with valid data	1. Open Gmail 2. Enter UserID 3. Enter password 4. Submit	UserID = abc@gmail.com Password = 123@abc	User should login	As expected, user Logged in	Pass
TU02	Check customer login with invalid data	1. Open Gmail 2. Enter UserID 3. Enter password 4. Submit	UserID = abc@gmail.com Password = abc@123	User should not login	As expected, user did not login	Pass

→ Conclusion

Thus, we successfully learned how to write various test cases in a specified format for better understanding & documentation, while using Gmail login Page as the test scenario.

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Assignment-2

→ Title

TEST cases for social Media Application

→ Objectives

- To teach students the process of writing effective test cases that cover different scenarios, including edge cases.
- To instill the importance of software testing in ensuring the quality & reliability of applications.

→ Problem statement

Write Test Cases in excel sheet for social Media application or website

→ Outcomes

- Students will be able to create detailed & comprehensive test cases for any software application
- Students will be able to communicate their test cases effectively, making it easier for developers to understand and implement the necessary fixes.

→ Software & Hardware Requirements

- Software

64-bit open source linux or its derivative, Microsoft Excel or its linux equivalent.

- Hardware

Stable Internet connection (atleast 2 Mbps)

→ Theory

To test the software, the software testing provides a particular format, called a test case.

→ Parameters of a Test Case

i) Test Case ID
It is a unique identifier assigned to every single condition in a test case. eg. TV01.

ii) Test Case Description
It is the condition required to be checked for a given software, like checking if the 'numbers only' validation is working or not for the age input box.

iii) Test Steps
These are the steps to be performed for checking of the condition.

iv) Test Data
The inputs to be taken while checking the conditions

v) Expected Result
The output which should be expected at the end of the test, granted everything goes right.

vi) Actual Result
The output that is actually found at the end of the test. It may or may not be equal to the expected result.

vii) Pass/Fail (Status)
If the actual result is same as the expected results, the test is declared successful & the status is 'Pass' and if the actual result is not as expected, the test is unsuccessful and its status is 'Fail'.

- We can also provide additional parameters such as Comments, Environment Information, Test Priority, Prerequisites, Tester Name, Test scenario, etc.

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→ Difference between Test Plan & Test Strategy

Sr. No.	Test Plan	Sr. No.	Test strategy
1.	A test plan for software projects outlines the purpose, methodology, goals, and emphasis of a software testing effort.	1.	A test strategy is a collection of rules that defines test execution and clarifies design of the test.
2.	Test Plan elements include test plan id, features to be tested, test methodologies, test tasks, features pass or fail criteria, deliverables, etc.	2.	Test strategy elements include objectives & scope, documentation formats, team reporting structures, client communication strategies, etc
3.	A testing manager or lead executes a test plan	3.	A project manager employs the test strategy
4.	It tells the story of specification	4.	Test strategy describes the standard methods
5.	Test Plan can change	5.	It cannot be changed.
6.	A test plan exists individually	6.	Test strategy is often found as a section of a plan
7.	It is defined at project level. It is done to identify immediate risks.	7.	It is set at organizational level and can be used by multiple projects. It is a long-term strategy.

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→ Why do we need Test cases?

Test cases are one of the most important aspect of software engineering, as they define how the testing would be carried out. There are many advantages of writing test cases -

1. To check whether the software meets customer expectations
Test cases help to check if a particular module/ software is meeting the specified requirement or not.
2. To check software consistency with conditions
Test cases determine if a particular module/ software works with a given set of conditions.
3. Narrow down software updates
Test cases help to narrow down the software needs & required updates.
4. Better Test Coverage
Test cases help to make sure that all possible scenarios are covered and documented.
5. Helpful during maintenance
Test cases are detailed which makes them helpful during the maintenance phase.

→ Conclusion

Thus, we successfully learned the importance of test cases & how to write test cases for a social media application.

DYP

STQA
Assignment-3

→ Title

Defect Report for an Application

→ Objectives

- Learn the process and importance of documenting defects in software applications.
- Study the various stages a defect goes through from identification to resolution

→ Problem Statement

Create Defect report for any application or web application

→ Outcomes

- Students will be better prepared for roles in software testing and quality assurance in professional settings.
- Ability to create comprehensive and accurate defect reports for any application or web application

→ Software & Hardware Requirements

- Software

64-bit Open source linux or its derivative, Word, Microsoft Excel or its Linux equivalent.

- Hardware

Stable Internet connection (atleast 2 Mbps)

→ Theory

Defect is the difference observed between expected & actual results which are noted post the event of testing a software.

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- A defect is found out after the product goes into production. So, If a developer finds an issue and corrects it by himself in the development phase then it is called a defect.

→ Classification of Defects

The defects are classified according to its impact on the functionality of the application. The different types of defects are-

i) Critical

The defects termed as critical need immediate attention & treatment. A critical defect directly affects the essential functionality, which may affect a software product on a large scale, such as failure of a feature or the whole system, etc.

ii) Major

These are responsible for affecting core & major functionalities. They don't lead to complete failure but may bring several major functions to rest.

iii) Minor

These defects produce minor impact, and do not have any significant influence on a software product. The results of these defects may be seen in the product's working, however, it does not stop users from executing their tasks, which can be carried out using different alternatives.

iv) Trivial

These types of defects have no impact on the working of a product, and sometimes, they are ignored and skipped, such as spelling or grammatical mistakes.

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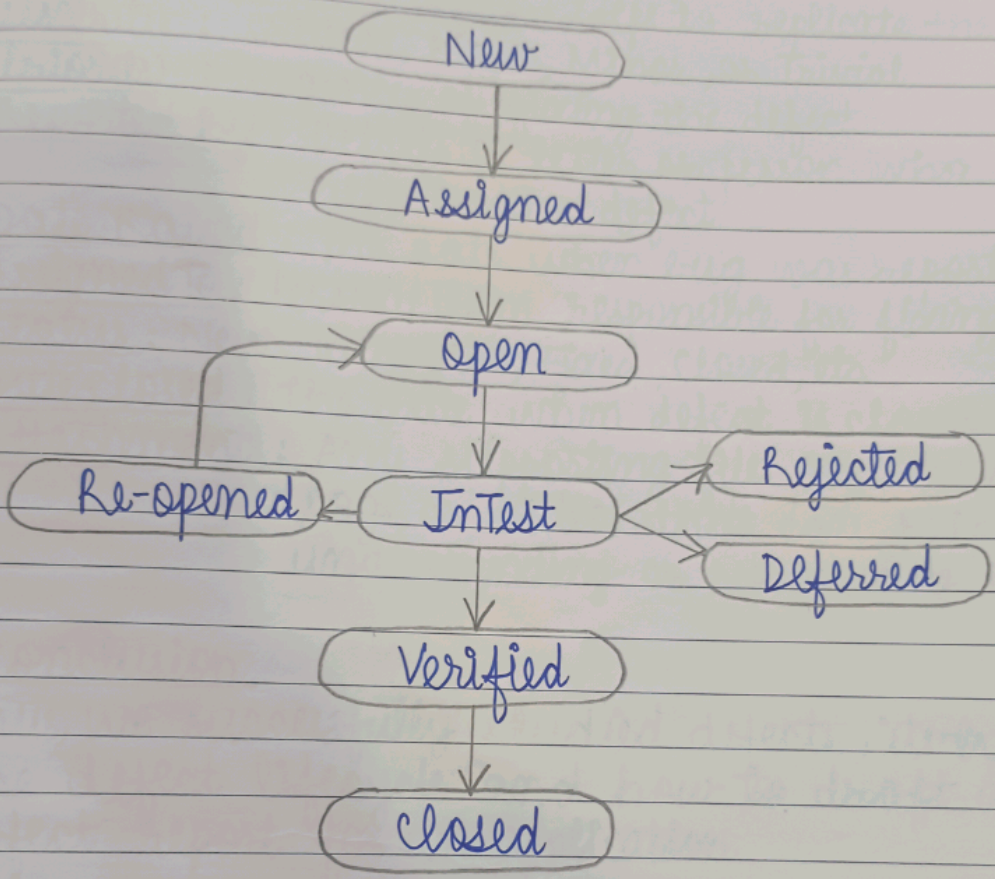
→ Defect life cycle

A defect life cycle is the movement of defect in different stages of its lifetime.



→ Defect Report

A defect report is a document that identifies and describes a problem found in the software product.

- The various elements of a defect report are-

1. Defect ID : A unique identifier for the defect, often auto-generated by the defect tracking system.

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2. Defect Description : A brief & clear description of the defect.
3. Product Version : The version of the software in which defect was found.
4. Steps to Reproduce Defect : Steps to replicate the defect.
5. Severity : Critical, Major, Minor, or Trivial
6. Priority : Urgency of fixing the defect
7. Reported By : Details of tester or person who reported the defect
8. Date Raised : The date when bug was reported.
9. Assigned To : Person/Team responsible for fixing
10. Status : New, In Progress, Fixed, Closed, etc.
11. Date closed : The date when defect is closed.
12. Attachments : Any supporting files like screenshots, logs, or videos that can help in understanding or reproducing the defect.

→ Conclusion

Thus, we successfully studied defects, its types, the defect lifecycle and how to draft a defect report for an application.

STQA

Assignment - 4

→ Title

Installation of selenium Grid, WebDriver & Java Eclipse

→ Objectives

- Gain hands-on experience in installing and configuring selenium grid to manage distributed test execution.
- Understand the functionality of selenium webdriver & its role in browser automation.
- Install & configure Java Eclipse, an IDE for Java programming.

→ Problem statement

Installation of selenium grid, selenium webdriver and Java Eclipse (automation tools).

→ Outcomes

- Students will enhance their problem-solving skills by troubleshooting and resolving issues during the setup and execution of automation tools.
- Students will learn to integrate various automation tools, facilitating smoother and more efficient automated testing processes in their future projects.

→ Software & Hardware Requirements

- Software

64-bit Open source linux or its derivative

- Hardware

Stable internet connection (atleast 2MBps)

DYP

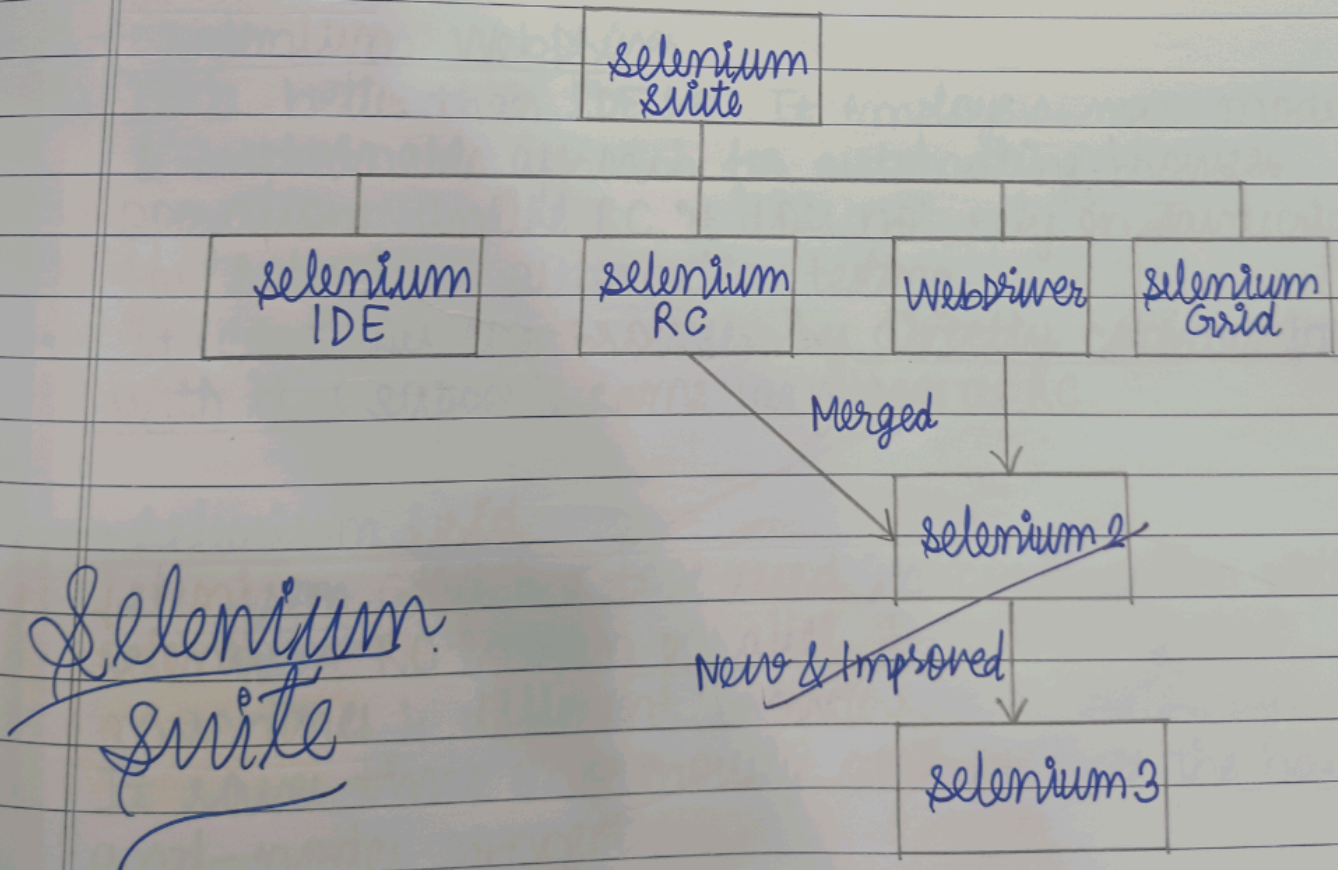
→ Theory

Selenium

Selenium is a free, open-source automated testing framework used to validate web applications across different browsers and platforms.

- We can use multiple programming languages like Java, C#, and Python to create selenium test scripts.
- Selenium software is not just a tool, but a suite of software, each piece catering to the different selenium QA testing needs of an organization.
- The equipment list is -

1. Selenium Integrated Development Environment (IDE)
2. Selenium Remote Control (RC)
3. WebDriver
4. Selenium Grid



Selenium Suite

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→ Selenium IDE

The selenium IDE is the simplest framework in the selenium suite & one of the easiest to learn.

- This is a firefox plugin that you can install as easily as other plugins
- However, due to its simplicity, it should only be used as a prototyping tool.

→ Selenium RC

Selenium RC was the flagship testing framework of the whole selenium project for a long time.

- This is the first automated web testing tool that allowed users to use a preferred programming language.
- As of version 2.25.0, RC supports Java, C#, PHP, Python, Perl, Ruby.

→ Selenium WebDriver

It is better than IDE & RC. It employs a more modern & sustainable approach for automating browser operations. Unlike RC, it does not rely on Javascript for selenium automation testing.

- It controls the browser by directly communicating with it & supports same languages as RC.

→ Selenium Grid

Selenium Grid is a tool used in conjunction with selenium RC to run parallel tests on different machines & different browsers simultaneously.

- It saves time enormously and utilizes the hub-and-nodes-concept.

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- The hub acts as a central source of selenium commands to each node connected to it.

→ steps to install selenium

1. Open a terminal & update your package list with 'sudo apt update'
2. Ensure Python and pip are installed.
3. Install selenium using 'pip3 install selenium'
4. Install Geckodriver. It is required for selenium to interact with firefox browser.
   'sudo apt install firefox-geckodriver'
5. Verify installation by running a simple code.
6. Handle any dependencies.

Java Eclipse

Eclipse is an IDE for developing applications using the Java programming language & others such as C/C++, Python, Perl, Ruby, etc.

- It can be used to develop rich client applications, IDEs, and other tools.

→ Steps to install Java Eclipse

1. Open a terminal and update your package list.
2. Eclipse requires a Java Development Kit (JDK) to run. So, install OpenJDK
   'sudo apt install openjdk-17-jdk'
3. Download Eclipse installer. Use 'wget'.
4. Extract the tar.gz file
5. Navigate to extracted directory & run the eclipse installer
6. select the Eclipse IDE for Java Developers
7. Choose the installation folder

8. Click the 'install' button & accept the license agreements.
9. Launch Eclipse
'~/eclipse/java-<version>/eclipse/eclipse'
10. Create a Desktop shortcut for future use.
'sudo nano /usr/share/applications/eclipse.desktop'

→ Conclusion

Thus, we have successfully studied and installed the automated testing tools like selenium grid & webdriver and also the IDE, Java Eclipse.

STQA
Assignment-5

→ Title
Software Requirement specification

→ Objectives
- learn the structure and components of a Software Requirement specification (SRS) document.
- Practice translating stakeholder needs into detailed and clear technical specifications.

→ Problem statement
Prepare software Requirement specification (SRS) for any project or problem statement.

→ Outcomes
- Ability to create a well-structured SRS document for any project or problem statement.
- Ability to communicate technical specifications to both technical and non-technical stakeholders.

→ Software & Hardware Requirements
- Software
64-bit open source linux or its derivative, Microsoft Word, Excel or its Linux equivalent.
- Hardware
Stable Internet connection (atleast 2MBps)

→ Theory
Software Requirement specification (SRS)
An SRS is a detailed description of the intended purpose, features, functionality, and other aspects of a software application.

→ Features of a good SRS Document

1. Correct
It should be correct, i.e. according to the needs of the customer
2. complete
All the software requirements should be mentioned in SRS.
3. Consistent
It should be consistent from the beginning so end users can understand easily & there must not be any contradicting requirements
4. Unambiguous
There should only be one interpretation.
5. Modifiable
All the requirements should be modifiable.
6. Verifiable
The requirements are verified by experts & testers
7. Traceable
Each requirement should have its own identity.
8. Testable
It should be capable of being tested.
9. Clear
The requirements should be clearly defined
10. Accurate
The requirements should have accuracy.

→ Importance of SRS

i) Clarity & Understanding, Risk Mitigation
(ii) Guidance for Development Teams
(iii) Facilitates software Testing & Quality Assurance
(iv) Client & Stakeholder satisfaction
(v) Forms the basis for Project Planning

SRS Document Format

1. Introduction
   - The purpose is defined which clarifies the purpose of the document and its intended audience.
   - The scope is defined
   - Definitions, Acronyms, & abbreviations
   - References
   - Overview
2. Overall Description
   - Product perspective, which is context or environment where product will operate
   - Product functions, user characteristics, constraints, assumptions, dependencies.
3. Specific Requirements
   - 3.1. Functional Requirements
     Inputs, outputs, behaviour for each function
   - 3.2. Interface Requirements
     User, hardware, software, communication interfaces
   - 3.3. Performance Requirements
     Speed, responsiveness, availability, scalability, design constraints of industry standards
   - 3.4. Security Requirements
     Data protection, Access control, Authentication
   - 3.5. Safety Requirements
     Safety of system & users.
   - 3.6. Quality Assurance Requirements
     Reliability, Portability, Usability, Maintainability
4. External Interface Requirements
   User, Software, Hardware, Communication interfaces
5. System Features
   Feature 1, 2... N

6. Non-functional Requirements

6.1. Business Requirements
Company policies, constraints, deadline, budget

6.2. User Requirements
Aesthetics, non-technical requirements

6.3. Documentation Requirements
Formats, Legacy documents

6.4. Performance, security, usability, reliability, maintainability, portability requirements.

7. Miscellaneous Requirements
legal, Regulatory, Environmental Requirements

8. Appendices
supporting information, Glossary, Data Models, Flow Diagrams, or additional resources.

→ Conclusion
Thus, we successfully studied the format of a software requirement specification document & its features & characteristics.